

Hierarchical Clustering and Anomaly Detection via DBSCAN

Albert Y. Kim

Introduction

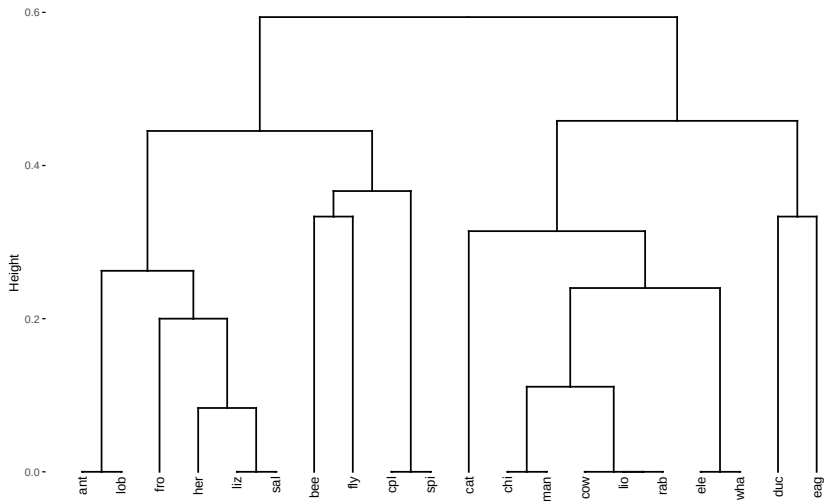
- ▶ Unsupervised methods identify patterns in the data without specifying an outcome measure
 - ▶ *k*-means clustering finds groupings of similar data-points using *prototypes*
 - ▶ *k*-means is an example of **partitional clustering**, as it produces clusters without any overlap

Hierarchical Clustering (overview)

- ▶ **Hierarchical clustering** organizes data-points into a tree-like structure known as **dendogram** that stores a series of nested (overlapping) groupings
- ▶ There are two major types of hierarchical clustering algorithms:
 1. **Agglomerative** - each data-point begins as its own cluster and pairs of clusters are merged using a *linkage criterion*
 2. **Divisive** - all data-points begin in a single cluster that is recursively subdivided until each data-point is its own cluster
- ▶ We'll focus on agglomerative clustering since divisive clustering algorithms aren't currently offered in `sklearn`

Hierarchical Clustering (dendrogram example)

Dendrogram Example (agglomerative clustering of animals)



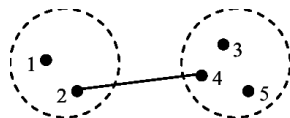
Agglomerative clustering - linkage

Three of the most popular ways to merge clusters are very straightforward:

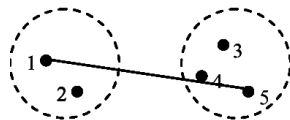
- ▶ *Single linkage* - find the minimum pairwise distance between data-points in different clusters and merge their clusters
- ▶ *Complete linkage* - find the maximum pairwise distance between data-points in each pairing of clusters and merge the two clusters with the smallest maximum
- ▶ *Average linkage* - find the average pairwise distances between points in each pairing of clusters and merge the two clusters with the smallest average distance

Agglomerative clustering - linkage

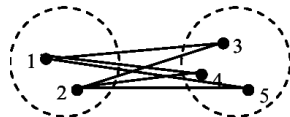
Diagram illustrating single, complete and average linkage:



$$d_{24}$$



$$d_{15}$$



$$\frac{d_{13}+d_{14}+d_{15}+d_{23}+d_{24}+d_{25}}{6}$$

Agglomerative clustering (Ward linkage)

The most common linkage criterion used in agglomerative clustering is *ward's linkage*, which minimizes Δ_{c_a, c_b} , the increase in sum of squared error accumulated by merging clusters:

- ▶ $\Delta_{c_a, c_b} = SSC_{c_a, c_b} - SSS_{c_a, c_b}$
 - ▶ $SSC_{c_a, c_b} = \sum_{i \in c_a \cup c_b} \|\mathbf{x}_i - \Psi_{c_a \cup c_b}\|^2$
 - ▶ $SSS_{c_a, c_b} = \sum_{i \in c_a} \|\mathbf{x}_i - \Psi_{c_a}\|^2 + \sum_{i \in c_b} \|\mathbf{x}_i - \Psi_{c_b}\|^2$

Note that c_a and c_b index the data-points in belonging to two different clusters and Ψ_{c_a} denotes the center of cluster c_a .

Choosing a linkage criterion

- ▶ *Single linkage* - tends to create large chains of one-at-a-time additions but can be good at identifying irregular patterns
- ▶ *Complete linkage* - robust to outliers and tends to favor similarly sized clusters at each “level” of the dendrogram
- ▶ *Average linkage* - lower variability than complete linkage but more impacted by outliers
- ▶ *Ward's linkage* - clusters tend to be the most compact (desirable) but the method is also the most computationally expensive

Agglomerative clustering vs. k -means

- ▶ k -means forms *spherical clusters* (thus it excels with the “blobs” data) but struggles to identify irregularly shaped clusters (ie: “moons” data)
 - ▶ Agglomerative clustering tends to be more flexible when applied to unusual data sets
- ▶ Agglomerative clustering can also better handle outliers and do not involve random initialization
- ▶ The main downside of agglomerative clustering is its computational burden when n is large

DBSCAN

- ▶ DBSCAN finds clusters using two parameters: a radius `eps`, and a minimum number of data-points `min_samples`
 - ▶ The algorithm surrounds each data-point with a hypersphere (or a circle in 2 dimensions)
- ▶ These hyperspheres are used to label each data-point as one of three types: *core points*, *border points*, and *noise*
 - ▶ Core points contain at least `min_samples` neighbors within their hypersphere
 - ▶ Border points contain at least 1 neighbor within their hypersphere
 - ▶ Noise points are at least `eps` away from any other data-point
- ▶ Cluster membership can then be determined by connected density regions

DBSCAN

The diagram below illustrates DBSCAN:

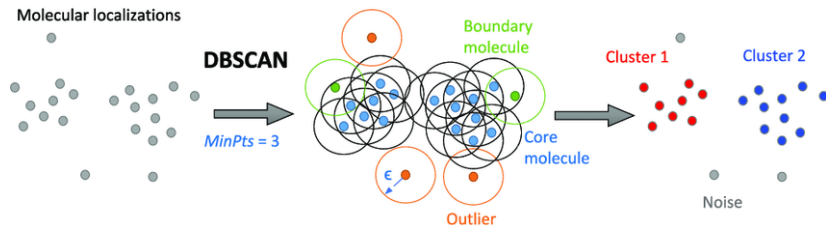


Image Source: [A Review of Super-Resolution Single-Molecule Localization Microscopy Cluster Analysis and Quantification_Methods](#)

Outlier (anomaly) detection

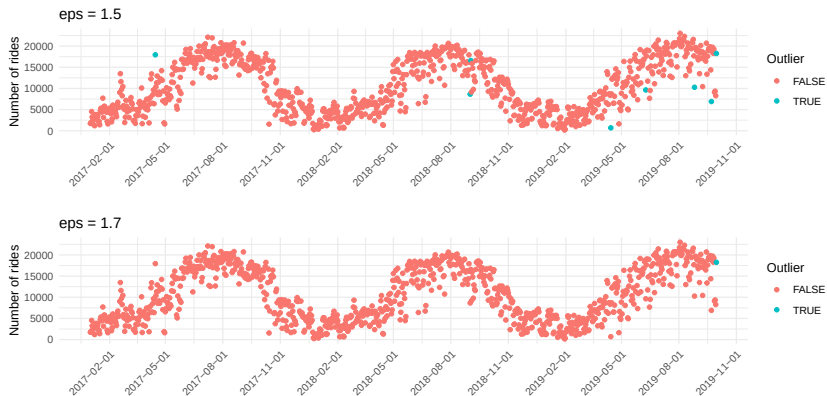
- ▶ In statistics, an outlier is a data-point that is significantly far from other observations
 - ▶ A simple example is the “3 sigma rule”, which will classify anything more than 3 standard deviations from the mean as an outlier
 - ▶ This amounts most extreme 0.3% of data-points under a normal model

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 - ▶ This amounts most extreme 0.3% of data-points under a normal model
- ▶ DBSCAN provides a flexible method of outlier detection governed by the `eps` hyperparameter
 - ▶ This can be set using domain-specific knowledge, or tuned so that a certain percentage of the data is classified as outliers

Example (Chicago Divvy bikeshare data)

Standardizing the number of rides only:



Example (Chicago Divvy bikeshare data)

Standardizing the number of rides and day:

